Lecture 41

Performance metrics

$\left\{ \underline{\text{imp}} \quad TPR = \frac{TP}{TP+FN} \; ; \; FPR = \frac{FP}{FP+TN} \right\}$

ROC & AUC curve. (mainly used in binary classif)

[Threshold values: 0, 0.2, 0.4, 0.6, 0.8, 1]

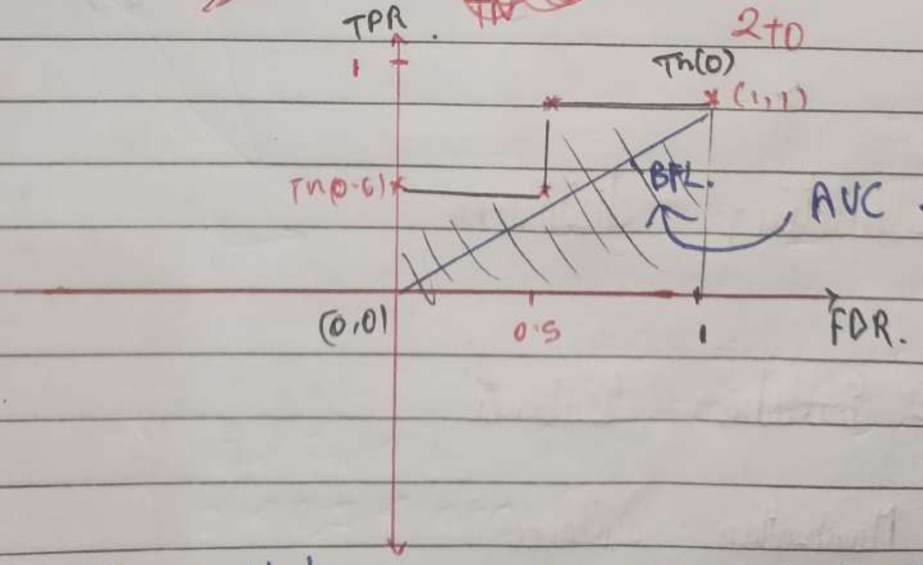
y	$\hat{y}$ prob	$\hat{y}$(0) output predicti	
1	0.8	1	- TP
0	0.96	1	FP.
1	0.4	1	- TP
1	0.3	1	- TP
0	0.2	1	FP (∵ i/p = 0,
1	0.7	1	- TP.

FN ⇒ when orig. val = 1 & o/p is 0

1: TPR (True Positive Rate) = n(TP)/(TP+FN)

$= \frac{4}{4+0} \Rightarrow \frac{4}{4} \Rightarrow 1$

1: FPR = $\frac{FP}{FP+TN} \Rightarrow \frac{2}{2+0} = 1$



AUC should be higher for a good pred model

A good model should have: less FPR & high TPR.